

VALLEY ANGLESEY, United Kingdom

WMO: 033020

Lat: 53.2525N

Lon: 4.5364W

Elev: 37

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.4	31.7	22.1	16.6	33.9	24.7	18.8	35.1	44.9	48.0	41.3	47.3	6.2	70	0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.2	73.1	62.8	69.3	60.9	66.4	59.7	64.0	70.1	62.5	66.6	61.2	64.5	9.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.9	82.9	65.3	60.8	79.7	63.8	59.8	76.9	62.9	29.2	70.5	28.1	66.8	27.3	64.6	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
38.8	33.7	30.1	DB	25.5	80.3	3.2	4.9	23.1	83.8	21.2	86.7	19.4	89.5	17.1	93.0
			WB	23.9	67.9	3.2	2.6	21.6	69.8	19.7	71.3	17.9	72.8	15.6	74.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	43.1	42.8	44.5	47.8	52.7	57.0	60.1	60.4	57.6	53.3	48.2	44.4	(6)
	DBStd	7.62	4.82	4.36	4.11	3.99	4.12	3.68	3.42	2.99	3.21	4.15	4.51	5.40	(7)
	HDD50	974	215	203	173	85	16	0	0	0	0	17	85	180	(8)
	HDD65	5112	679	623	634	515	381	243	159	149	222	362	505	640	(9)
	CDD50	1357	1	1	3	20	101	211	314	321	229	120	30	6	(10)
	CDD65	17	0	0	0	0	0	3	8	5	1	0	0	0	(11)
	CDH74	96	0	0	0	0	4	18	45	26	3	0	0	0	(12)
	CDH80	13	0	0	0	0	0	1	8	4	0	0	0	0	(13)
(14) Wind	WSAvg	14.3	17.0	16.1	14.5	13.0	13.2	12.5	12.0	12.4	13.4	15.4	15.7	16.8	(14)
(15) Precipitation	PrecAvg	33.3	3.1	2.3	2.4	2.1	2.0	2.1	2.2	2.8	2.9	3.8	4.1	3.7	(15)
	PrecMax	41.5	6.3	6.3	6.0	4.1	5.0	5.1	5.0	5.6	6.3	8.2	8.7	10.0	(16)
	PrecMin	24.6	0.3	0.3	0.5	0.3	0.4	0.4	0.4	0.4	0.1	0.7	1.1	0.7	(17)
	PrecStd	3.9	1.4	1.2	1.1	1.0	1.1	1.1	1.1	1.1	1.5	1.9	1.6	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	53.4	60.4	66.6	72.3	76.8	79.1	77.5	71.8	65.1	58.9	54.9	(19)
		MCWB	50.5	47.6	51.5	55.8	61.4	62.7	65.2	65.2	62.4	58.2	55.7	52.8	(20)
	2%	DB	51.4	50.8	54.0	59.4	66.5	70.2	73.2	70.8	67.0	61.8	57.0	53.7	(21)
		MCWB	49.8	48.5	48.7	51.8	58.3	60.4	63.1	62.5	59.7	57.2	54.9	52.1	(22)
	5%	DB	50.4	49.5	51.2	55.6	62.3	66.0	68.7	67.1	64.5	60.3	55.7	52.6	(23)
		MCWB	48.6	47.6	47.8	49.6	55.6	58.2	61.2	60.9	59.2	56.9	53.4	50.8	(24)
	10%	DB	49.4	48.6	49.7	53.4	59.0	63.0	65.5	64.7	62.7	59.2	54.5	51.4	(25)
		MCWB	47.6	46.8	47.0	49.1	53.5	57.2	60.0	59.7	58.3	56.4	52.2	49.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	50.4	52.5	55.8	62.6	64.2	67.1	66.5	64.2	60.3	56.9	53.8	(27)
		MCDB	51.9	51.1	57.6	64.8	70.3	72.5	76.2	74.6	69.6	62.7	57.7	54.3	(28)
	2%	WB	50.4	49.2	50.0	52.8	58.9	61.4	64.0	63.8	61.8	59.1	55.5	52.6	(29)
		MCDB	51.0	50.1	52.3	57.6	65.6	68.2	70.7	68.4	64.6	60.6	56.7	53.3	(30)
	5%	WB	49.3	48.2	48.8	51.1	56.4	59.4	62.3	62.3	60.7	58.0	54.1	51.3	(31)
		MCDB	50.1	49.1	50.5	54.2	60.9	64.2	66.9	65.0	62.9	59.7	55.4	52.3	(32)
	10%	WB	48.2	47.3	47.8	49.9	54.6	58.0	60.8	61.2	59.6	56.9	52.7	50.1	(33)
		MCDB	49.3	48.4	49.3	52.4	57.9	61.9	64.3	63.6	61.9	58.8	54.3	51.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.8	8.9	9.7	10.3	10.0	9.5	9.2	9.4	8.5	7.4	7.4	(35)
		MCDBR	6.3	7.0	10.2	13.3	15.9	16.2	15.8	13.1	12.0	9.0	7.1	6.8	(36)
	5% WB	MCWBR	7.3	6.5	7.4	8.0	9.7	8.9	8.1	7.4	7.5	7.0	7.3	7.8	(37)
		MCWBR	6.4	6.1	8.8	10.7	13.9	13.0	13.1	10.3	9.3	7.6	6.6	6.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.334	0.345	0.371	0.389	0.394	0.386	0.389	0.374	0.366	0.353	0.330	0.322	(40)
		taud	2.308	2.374	2.336	2.316	2.355	2.406	2.402	2.458	2.471	2.448	2.373	2.284	(41)
	Ebn at Noon		199	239	257	267	271	275	271	270	257	233	200	177	(42)
		Edh at Noon	20	25	31	36	36	35	35	31	27	23	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		207	435	828	1308	1663	1769	1611	1350	962	533	256	152	(44)
	RadStd		19	37	60	88	116	148	136	91	59	47	25	15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page